

Article

Research on Children's Body Proportions: Determining the Canon of Head Length to Total Body Height on the Example of Children Aged 2 to 15 Years

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Abstract: Proportions and canons of the human body have always been an area of research mainly through art, architecture, or construction, and today, they have a significant application in product design. Research confirms that body height in most cases corresponds to the canon (head–body ratio) of 7.5 to 8 head lengths. This paper investigates the ratio of the head length (HL) to the total body height (BH, stature) of kindergarten and school-aged children, aiming to define the children's canon inspired by the idea of the harmonic circle theory and the biomechanical model. The data were collected from 1307 children (male 676, female 631) aged 2 to 16 years in the cities of Zagreb (Croatia), Sofia (Bulgaria), and Skopje (North Macedonia). A generalized ESD test (alpha-level 0.10) and Turkey's 1977 test were used in order to detect outliers in distributions of heights and in the distribution of ratios. Statistical significance was set at 0.05, all *p* values were two-sided, and the MedCalc statistical tool (version 20.110) was used. The results confirm that canonical changes follow the historical research of artists throughout the centuries, but that they change according to contemporary secular trends in children's growth and cover HL/BH canons from 5.59 and 5.72 (2-year-old girls and boys) to 7.50 and 7.60 (15-year-old boys and girls) depending on age and gender. HL/BH ratio was significantly higher among female examinees in all age groups where difference was significant (Student's *t* test, *p* < 0.02). In conclusion, such a calculation based on the canon is important for interdisciplinary professions. Creating an anthropological–biomechanical model based on canons, instead of time-consuming measurement, could significantly simplify the long-term collection of anthropometric data used for designing children's products. Future detailed research is proposed.

Keywords: body proportions; canons; head–body ratio; children anthropometry; children's product design



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1. Introduction

The perception of human proportions has changed through different epochs [1]. Artists, philosophers, and contemporary experts and scientists have dealt with harmony, canons, and proportions since ancient times [2,3]. Canon is considered as the body of rules, principles, or standards accepted as axiomatic and universally binding in a religion, art or philosophy [4]. The study of changes in proportions and the determination of the existing canons of the human body, whether it is about children or the elderly population, helps artists in drawing body parts [5,6] and scientists in anthropometry and ergonomics to determine the main anthropometric characteristics in dynamic anthropometry and biomechanical analyses [6,7], while it helps designers in the design of products [8–10].

The ratio of the head to the total height of an adult has been known in the literature for a long time; however, very little research has been conducted on this ratio in preschool and school children [11–13]. In the available (but at the same time outdated) literature, anthropometry for adults is relatively well covered [14,15], while for children, especially in terms of proportions, it is still not sufficient [6,16,17]. The available data on the head length and growth ratio in children are older than 20 years [6], so the question arises, are they still valid today? Due to the secular trends that appear in anthropometry, new data on the canon (ratio) would be extremely useful and could serve to reduce time and save financial resources when measuring children and determining their anthropometers, but also when designing furniture and other products for children.

The current research started from the idea to confirm the historical canons that have been studied by artists and engineers for centuries and improve the understanding of changes in body proportions during children's development that could be correlated with the canon and, at the same time, to indicate the necessity of updating the existing anthropometric data in order to ensure a more effective and adaptive design of children's products, especially furniture, as well as to consider the necessity of future interdisciplinary research.

The data used in this paper were taken from the author's published doctoral thesis [11,12], where the main focus was on establishing the (in)compatibility of furniture in educational institutions with the dimensions of children. The results presented in this paper have not been compared anywhere or previously published in this way.

1.1. Previous Studies of the Canons and Proportions of the Human Body

The canon of proportions was first used in Egypt [18,19]. One of the earliest documents on the proportions of the human body was written by Marcus Vitruvius Polio, a Roman architect and writer from the 1st century AD [20]. He begins his Ten Books on Architecture with the recommendation that temples should be constructed on the analogy of a beautifully shaped human body in which there is harmony of all parts [21]. The Vitruvius canon, where the head is $1/8$ of the body is still most often used today, especially if body proportions do not require a more precise division [11].

During the Renaissance, Leonardo da Vinci took the Vitruvian model to which he added line and triangular diagrams that show how adjacent parts of the body contain proportions that range between the golden ratio and the Pythagorean rule [22].

Albercht Dürer's Four Books on Human Proportions published in 1594 was influenced by Leonardo da Vinci, Marcus Vitruvius, and other significant thinkers [21]. Before Dürer's drawings, there was only one absolute form of beauty based on ideal proportions that were determined by Vitruvius [23]. Dürer, studying and illustrating human harmonic scales and the relationships between drawings of the body of a child and an adult man, based body proportions on strict mathematical modules [1], where he used the entire head length (HL) in a ratio of $1/8$ of the body height (BH) as a unit of measurement.

The proportions of the human body were intensively studied in detail at the end of the 19th century by the French anatomist and painter Richer [24], who established the canonical rule of 7.5 HL to the body height (BH) of an adult.

Barcsay [5] analyzes the canons of the child's body with regard to the observed age, where it is noted that the differences in body proportions are of great importance depending on the age of the child. Research by Muftić [25] showed similar results compared to Barcsay [5]. The most obvious changes are the proportions between the HL and the height of the entire body (BH), according to certain proven regularities.

Muftić [25] tried to define these laws with the theory of the harmonic circle and the harmonic analysis of the anthropometry of the child's body, and to determine the appropriate biomechanical model by determining the canons for a particular age group. He developed a method of harmonic anthropometric analysis related to the standing height of a person [26]. Linear connections between anthropological proportions were determined using the canon of 8 HL associated with the so-called harmonic circle created by Zederbauer in 1917. Using this idea, a circle was drawn, whose diameter was on a scale equal to the

height of a man and a grid that represents the boundaries of the contour of a man in a standing body position. Characteristic points on the body representing joints or end points of the contour were determined. The coordinates of the points were determined as a function of the total height of the person and $1/8$ HL [7].

All the mentioned canons apply to the population of mature adults. If one wants to apply the same canon to the younger population—children—it is evident that the proportions of the head, and thus of the whole body, starting from a newborn to an adolescent of 18 years, change significantly and continuously (Figure 1). Body measurements of children proved this during yearly systematic examinations [27].

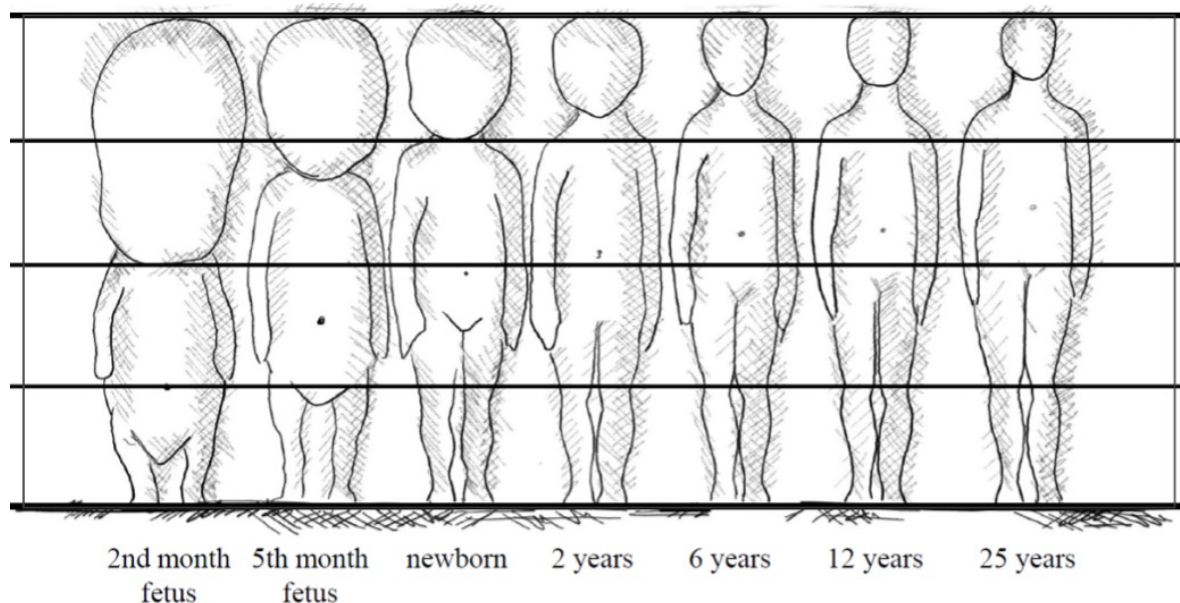


Figure 1. Proportion of the head of a newborn and grown man. (@ Domljan, 2024 adapted from [27]).

Studying the relationships of body parts of healthy people of younger and older age, it was observed that there are defined canons that change properly according to the growth and development of the human body at a certain age [6]. As the first step in children's dynamic anthropometry, the two main anthropometric characteristics—standing height and current body mass (weight)—of children are most often measured as functions of canonical and harmonic dependence. The result showed that the newborn belongs to the so-called canon 4 HL, while a four-year-old child has canon 5.5 HL [6].

It is interesting to compare the obtained mathematical calculation [7,25] with children's drawings [5]. The indicated proportions in the drawings of children from newborns to five- and ten-year-olds up to a fourteen-year-old boy show the same ratio of head length to body as in the mathematical calculation. The height of a newborn barely exceeds 4 HL; in five-year-olds, the body height slightly exceeds 5.5 HL, while in ten-year-olds, the height is slightly higher than 6 HL, and in fourteen-year-olds, the body height has not yet reached 7 HL [5].

According to the growth curve, the human body reaches its maximum body height between the ages of 17 and 25, which means that body height in most cases corresponds to the canon of 7.5 to 8 HL, which was also discussed by Vitruvius, Durer, and Leonardo.

From the above, it can be observed that between the ages of 6 and 15, a child intensively changes its body proportions in relation to the length of its head. But is there a functional dependence of the change of the associated canon depending on the age of the child?

Muftić went a step further [6]. Measuring children between the ages of 3 and 18, he observed that the functional curves for standing height, whose equations are interpolated curves of the second order in the age range of two to eight years, statistically significantly differ from each other only after the eighth year of life. After that, and especially after the

tenth year, differences appear that should be respected. This diversity was the impetus for the harmonic analysis of canon changes to be carried out specifically for boys and specifically for girls. A similar result was obtained by calculating the mean values of body mass with an age difference: for the measured male and female subjects, the statistical distribution of body mass does not differ significantly until the age of ten, only after that do the differences that need to be considered begin to appear [7].

1.2. The Aim of the Study

This study aims to analyze the ratios of head length and body height of children aged 2 to 16 years (concerning the gender of the children) and assess the applicability of the canons across different age groups, providing foundational insights that can inform the design of age-appropriate products suitable to children's evolving physical proportions. The idea behind this study is to confirm the children's canon and provide the basis for the creation of a comprehensive anthropological–biomechanical model that not only validates existing canons but also explores potential correlations with other anthropometric variables critical for ergonomic and biomechanical analyses.

The hypotheses presented in this paper are

- The canons between 5 HL and 7.5 HL for children of kindergarten and primary school age obtained throughout history are still valid today.
- It is possible to establish a unique canon for children of kindergarten and primary school age (2 to 15 years), which would be used as an aid in the design of various products for children.
- It is possible to determine the grounds for creating an anthropological–biomechanical model, which can be used to determine other anthropometric variables.

2. Materials and Methods

2.1. Cities/Countries

Data were collected in three capital cities of three countries: Zagreb in Croatia; Sofia in Bulgaria; and Skopje in North Macedonia. Observed cities/countries are a part of Balkan, a peninsular region of southeastern Europe, and have similar educational processes and starting ages of children in certain educational levels of schooling. They are also similar in size across children of certain ages, which enabled further comparison.

2.2. Participants

A cross-sectional study was conducted to measure the anthropometric variables of preschool and school children aged 2–16 years. All children who attended educational institutions on the days of measurement were European white race of the Slavic ethnic group.

The respondents were healthy children who did not show symptoms of chronic back pain, neck pain or any form of malformation in the musculoskeletal system, and who had moderate levels of physical activity, healthy eating habits, and no acute medical and family histories or unbalanced environmental exposures. None of the participants had previously participated in this type of study.

2.2.1. Preschool Children

Anthropometric measurements of preschool children included a total of 747 children, of which there were 383 boys and 364 girls [12]. Data on the dimensions of kindergarten children were taken from kindergartens in all three countries, where each child belonged to a specific age group. Five main groups were selected (groups I–V, Table 1). The youngest child at the time of the measurement was 2 years and 2 months old and the oldest was 7 years and 6 months old. The research was conducted in the period from January 2017 to January 2020.

Table 1. Distribution of preschool and school children by age groups and polygons.

Group/Class	Children's Age	Cities/Countries (Number of Children per Location)	Number (Total) (F)	Frequency (%) (Fr)
Preschool children				
I	≥2–<3 years	Skopje (8); Sofia (2); Zagreb (11)	21	2.8
II	≥3–<4 years	Skopje (30); Sofia (46); Zagreb (58)	134	17.9
III	≥4–<5 years	Skopje (71); Sofia (71); Zagreb (66)	208	27.8
IV	≥5–<6 years	Skopje (86); Sofia (62); Zagreb (62)	210	28.1
V	≥6–<7+ years *	Skopje (25); Sofia (89); Zagreb (60)	174	23.4
	N	Skopje (220); Sofia (270); Zagreb (257)	747	100.0
School children				
1	≥6.5–<7.5 years	Zagreb	51	9.4
1–2	≥7.5–<8.5 years	Zagreb	22	3.9
2–3	≥8.5–<9.5 years	Zagreb	73	13.0
3–4	≥9.5–<10.5 years	Zagreb	84	15.0
4–5	≥10.5–<11.5 years	Zagreb	88	15.7
5–6	≥11.5–<12.5 years	Zagreb	69	12.3
6–7	≥12.5–<13.5 years	Zagreb	64	11.4
7–8	≥13.5–<14.5 years	Zagreb	78	13.9
8	≥14.5–<16 years	Zagreb	30	5.3
	N	Zagreb	560	100.0
Total (all polygons)			1307	

Note. F = frequency; Fr = percentage of respondents, relative frequency * In the Republic of Macedonia, since 2008, according to the Law on Primary Education, children from 6 to 7 years enter the education process of primary schools, not the preschool process as before [28].

2.2.2. School Children

A total of 560 students, 298 boys and 262 girls, from first to eighth grade (Classes 1–8, Table 1) of elementary schools in the City of Zagreb (Croatia) participated in the anthropometric measurements in elementary schools [11]. The respondents were healthy children aged 6.5 to 16 years. The research was conducted in the period from June 2004 to June 2005.

Since children of a certain age do not necessarily attend the same group or classes, a distribution by age and gender was made in Table 2. The age of the children was taken as the starting point for further analysis of the canon. A total of 1307 examinees with ages from 2 to 16 were divided into 14 groups (i–ivx) based on age, and data analysis was performed for each group separately, as all three examined values depend on the age of a person. There was a similar ratio for both genders.

Table 2. Demographic characteristics of all children by age and gender.

	Group	Children's Age	N (Total Number of Children)	Male	Female	Frequency (Fr) (%)
Age	i	≥2–<3	21	11	10	1.6
	ii	≥3–<4	134	68	66	10.3
	iii	≥4–<5	208	96	112	15.9
	iv	≥5–<6	210	119	91	16.1
	v	≥6–<7	174	84	90	13.3
	vi	≥7–<8	57	35	22	4.4
	vii	≥8–<9	59	33	26	4.5
	viii	≥9–<10	73	44	29	5.6
	ix	≥10–<11	87	31	56	6.7
	x	≥11–<12	77	38	39	5.9

Table 2. Cont.

Group	Children's Age	N (Total Number of Children)	Male	Female	Frequency (Fr) (%)
xi	≥12–<13	66	37	29	5.0
xii	≥13–<14	71	35	36	5.4
xii	≥14–<15	62	38	24	4.7
ivx	≥15–<16	8	7	1	0.6
Total		1307	676 (51.7%)	631 (48.3%)	100.0

2.3. Anthropometric Variables Measured

For this paper, two main body variables from [11,12] were observed, according to [15] (Table 3).

Table 3. Anthropometric variables included in the research. (@ Domljan, 2024).

Stature (A)	The vertical distance from the vertex of the head (V) to the substrate on which the feet are placed. The head is positioned in the horizontal Frankfort Plane [29].	
Head length (B)	Vertical distance from the vertex of the head (V) to the top of the chin (gnathion, gn).	

The variables were measured in basic posture in the upright position of each child. The children were in a maximal upright position, with the head in a horizontal Frankfort Plane, the upper tonsils are aligned parallel to the body, facing the body. The lower tonsils are touched next to each other [29]. Two main instruments have been used during the measurements:

- The Altmeter was used for body height measurements, dimensions from 0 to 2000 mm [30];
- The Calliper was used for the head length measurements [31] (the shubler mehanická) 16090014//—from 0 to 600 mm, with calibration sheet “č. P540/16”, with an error margin of ± 0.02 mm.

Measurement and recording of variables were carried out separately for each child by two, always the same persons familiar with the measurement methodology, where one person measured the children and the other entered them in a pre-prepared form. During the measurement process, the children in all cities/countries were dressed in similar clothes: T-shirt and cotton pants or tights which they usually wear in a physical education class, without shoes. The time of measurement was from 9 a.m. to 13 p.m. during the working days (Mondays to Fridays).

2.4. Ethical Approval and Permission for Conducting the Research

Permissions to enter kindergartens and schools to conduct research were obtained in all locations/cities from the competent ministries, individual state or city offices, kindergarten and school principals, and parents/guardians whose children participated in the

research [11,12]. Additionally, at the polygons in Zagreb (Croatia), approval and confirmation that the research complies with the principles and rules of the Code of Ethics was given by the Ethics Committee of the University of Zagreb [32]. Informed consent was obtained from all research participants (principals, teachers, parents/guardians). All participants were informed about the aim of the study and possible withdrawal at any stage.

2.5. Statistical Methods

A generalized ESD test (alpha-level 0.10) and Turkey's 1977 method were used in order to detect outliers in distributions of heights and in distribution of ratios. Kolmogorov–Smirnov test was used to test normality of distribution, so data are presented with arithmetic mean and standard deviation and additionally with median, total range, and skewness coefficient.

Differences in proportions of categorical data were tested with Chi-square test, while differences in numerical data between two independent groups (regarding gender) were tested with Student's *t* test.

Statistical significance was set at 0.05, all *p* values were two-tailed, and statistical tool MedCalc (version 20.110) was used.

3. Results

The data were taken from 1307 children between the age of 2 and 16 years who participated in the research [11,12], with a similar ratio of both genders (males *N* = 676, 51.7%, Chi-square test: *p* = 0.38). Participants were divided into 14 groups by age, and data analysis was made for each group separately, as all three examined values (gender, ratio, and location but no comparison was made between cities) depend on the age of a child.

Distributions of body height and head length were first tested for outliers with two different tests for redundancy (in order to be more certain in outlier detection). The distribution of the body height variable does not have any outliers (generalized ESD test), neither outside nor far-out values, according to the Turkey's 1977 test. The distribution of the head length variable does not have outliers regarding the generalized ESD test, but has six outside values regarding Turkey's 1977 test and no far-out value. The distribution of the ratio of heights (body height divided by head length) also does not have outliers regarding the generalized ESD test, but has seven outside values regarding Turkey's 1977 test and no far-out value. Outliers should be removed in anthropometric values/measurements for each distribution; however, in total, it is only 12 (1.6%) examinees, so the authors decided to remove all of them from any kind of further analysis. There was no significant difference regarding gender between cities (Chi-Square test, *p* = 0.79), nor regarding gender between age (Chi-Square test, *p* = 0.19), but there was significant difference regarding age between cities (Chi-Square test, *p* < 0.001) as there were younger children in Skopje, Sofia, and Zagreb, but older ones only in Zagreb. After removing the identified outliers (e.g., children aged 16), the new distribution was tested for outliers again, and newly identified outliers were removed, if any. In the next step, the ratio between body height and head length was calculated using data cleaned of outliers. Identified outliers for each step and each age group are presented in the last three columns of the table, while the decrease in the number of examinees (second column in table) presents how many outliers were identified and removed (Table 4).

As outliers should be removed in anthropometric values/measurements for each distribution [15], in the last step, the obtained distribution of the ratio was also tested, and any identified outliers were also removed. The final result of the ratio between body height and head length for each age group is highlighted (bolded) in Table 4.

For some distributions of body height or head length, the skewness coefficient was high, but only before removing the outliers (and not in the distribution that was used for ratio calculation). However, some distributions remained not-normal-like (regarding Kolmogorov–Smirnov test) even after removing outliers, but the skewness coefficient was reduced greatly and differences between the mean and median were quite small. None of

the final ratio distributions are not-normal-like, and neither has high difference between mean and median, even for the last age group with very small sample size.

Table 4. Examined distributions before and after extracting outliers with finale ratio per age group.

Age Group	N		Mean (SD)	Median	Min–Max	Skewness	Turkey, 1977 Outside	Turkey, 1977 Far-Out	Generalized ESD Test
≥ 2 –<3	21	body	95.2 (3.9)	95.4	86.7–101.4	−0.506	none	none	none
		head	17.2 (0.9)	17.0	15.5–19.0	0.399	none	none	none
		ratio	5.56 (0.35)	5.61	4.73–6.11	−0.969	4.73	none	4.73, 4.81
		ratio	5.65 (0.35)	5.70	5.16–6.11	−0.221	none	none	none
≥ 3 –<4	134	body	101.8 (3.9)	101.5	89.7–112.3	−0.183	one 89.7	none	none
		head	17.5 (1.0)	17.5	15.6–22.0	0.885 *	20, 20	22	22
		body	101.9 (3.8)	101.5	92.2–112.3	−0.018	none	none	none
		head	17.5 (0.9)	17.5	15.6–19.6	0.281	none	none	none
	130	ratio	5.83 (0.31)	5.86	4.95–6.57	−0.395	4.95, 5.06, 5.11, 6.57	none	none
	126	ratio	5.84 (0.28)	5.86	5.19–6.43	−0.264	none	none	none
≥ 4 –<5	208	body	107.8 (4.7)	107.5	94.2–118.7	0.018	94.2	none	none
		head	18.0 (1.1)	17.8	15.3–21.5	0.459 *	21, 21.2, 21.5	none	none
		body	107.9 (4.6)	107.5	95.5–118.7	0.104	none	none	none
		head	17.9 (1.0)	17.8	15.3–20.5	0.196 *	15.3	none	none
	204	head	17.9 (0.9)	17.8	15.6–20.5	0.270 *	none	none	none
	203	ratio	6.00 (0.33)	5.99	5.25–6.95	0.241	6.95	none	none
	202	ratio	6.00 (0.32)	5.99	5.25–6.79	0.170	none	none	none
≥ 5 –<6	210	body	114.6 (5.1)	114.6	93.4–132.7	−0.089	129.9, 132.7, 22.1, 22.5, 22.5	93.4	93.4, 132.7
		head	18.7 (1.2)	18.6	16.0–22.5	0.494	102.5, 102.6	none	none
		body	114.5 (4.6)	114.5	102.5–126.3	−0.082	102.5, 102.6	none	none
		body	114.7 (4.5)	114.6	104.5–126.3	0.021	126.3	none	none
	204	body	114.6 (4.4)	114.6	104.5–124.6	−0.030	none	none	none
	207	head	18.6 (1.1)	18.6	16.0–21.8	0.244 *	none	none	none
	201	ratio	6.17 (0.35)	6.16	5.33–7.21	0.329	7.21	none	none
	200	ratio	6.17 (0.34)	6.16	5.33–7.03	0.254	none	none	none
≥ 6 –<7	174	body	122.8 (5.2)	122.6	110.3–138.0	0.185	136.5, 138	none	none
		head	19.4 (1.1)	19.3	16.6–25.5	1.313 *	22.5, 23, 23	25.5	23, 23, 25.5
		body	122.7 (5.0)	122.6	110.3–134.5	0.033	none	none	none
		head	19.3 (0.9)	19.3	16.6–22.0	0.117 *	none	none	none
	172	ratio	6.37 (0.32)	6.35	5.58–7.34	0.248	5.58, 7.29, 7.34	none	none
	168	ratio	6.37 (0.30)	6.35	5.66–7.12	0.105	7.11, 7.12	none	none
	163	ratio	6.36 (0.29)	6.35	5.66–6.97	−0.002	none	none	none
≥ 7 –<8	57	body	128.2 (6.3)	129.0	111.6–141.0	−0.278	none	none	none
		head	20.5 (2.3)	20.0	17.7–33.0	3.107 *	25	33	25, 33
		head	20.2 (1.4)	20.0	17.7–24.0	0.481	none	none	none
	55	ratio	6.36 (0.42)	6.41	5.17–7.42	−0.307	none	none	none
≥ 8 –<9	59	body	136.0 (5.9)	136.0	124.0–149.0	0.048	none	none	none
		head	21.3 (1.5)	21.0	18.0–26.0	0.283	18, 18, 24.5, 26	none	26
		head	21.3 (1.2)	21.0	19.0–24.0	0.039	none	none	none
	55	ratio	6.40 (0.39)	6.38	5.65–7.32	0.238	none	none	none
≥ 9 –<10	73	body	139.5 (5.9)	139.0	127.0–158.0	0.618 *	152.5, 155, 158	none	158
		head	21.1 (1.9)	21.0	17.0–29.0	0.915 *	none	29	29
		body	138.8 (5.0)	138.8	127.0–151.0	−0.011	none	none	none
		head	20.9 (1.7)	21.0	17.0–24.5	0.060 *	none	none	none
	69	ratio	6.68 (0.57)	6.65	5.59–8.41	0.698	8.32, 8.41	none	8.32, 8.41
	67	ratio	6.63 (0.50)	6.63	5.59–7.72	0.196	none	none	none
≥ 10 –<11	87	body	145.3 (6.2)	145.0	129.0–158.5	0.057	129	none	none
		head	21.3 (1.6)	21.5	18.0–25.5	0.149 *	25.5	none	none
		body	145.5 (6.0)	145.0	133.5–158.5	0.208	none	none	none
	86	head	21.3 (1.5)	21.3	18.0–25.0	0.016 *	none	none	none
	85	ratio	6.86 (0.51)	6.77	5.79–8.06	0.377	none	none	none
≥ 11 –<12	77	body	152.3 (8.2)	152.0	126.0–167.0	−0.279	126	none	126
		head	21.6 (1.8)	22.0	17.5–28.0	0.381 *	28	none	28
		body	152.6 (7.7)	152.0	139.0–167.0	0.059	none	none	none
		head	21.5 (1.6)	22.0	17.5–25.5	−0.146 *	none	none	none
	76	ratio	7.11 (0.49)	7.13	6.00–8.66	0.375	8.66	none	8.66
	74	ratio	7.09 (0.47)	7.12	6.00–8.32	0.082	none	none	none
≥ 12 –<13	66	body	158.5 (7.0)	157.8	139.0–171.5	−0.192	none	none	none
		head	21.4 (1.5)	21.5	18.0–24.0	−0.198 *	none	none	none
		ratio	7.44 (0.61)	7.31	6.32–8.95	0.456	none	none	none
≥ 13 –<14	71	body	163.9 (7.7)	164.0	144.5–183.0	−0.221	none	none	none
		head	22.0 (1.5)	22.0	19.0–26.5	0.355 *	26.5	none	none
	70	ratio	7.49 (0.43)	7.43	6.59–8.60	0.189	none	none	none
≥ 14 –<15	62	body	169.4 (7.8)	169.0	154.2–186.0	0.094	none	none	none
		head	22.3 (1.7)	22.5	18.0–26.0	−0.161	none	none	none
		ratio	7.62 (0.52)	7.54	6.76–8.90	0.189	none	none	none
≥ 15 –<16	8	body	170.3 (7.9)	168.0	162.0–183.5	0.768	none	none	none
		head	23.0 (2.5)	22.3	20.0–28.0	1.134	none	none	20, 21, 22.5, 24, 24.5, 28
		head	22.0 (–)	–	22.0–22.0	–	–	–	–
	2	ratio	7.48 (0.10) †	7.48	7.41–7.55	–	none	none	none

* Reject normality (Kolmogorov–Smirnov test); † sample too small.

The data from Table 4 show that body height and head length have an increasing trend up to the age of 8 (about 4 cm total change in head length from 2 to 8 years), and from 9 to 16 years, the length of the head remains the same or changes slightly (total increase of about 1 cm). The ratio between body height and head length was significantly higher among female examinees in all age groups where difference was significant (Student's *t* test, $p < 0.02$) (Figure 2). However, it seems that as the age of the examinees increases, the gender ratio decreases and is no longer significant (Table 5).

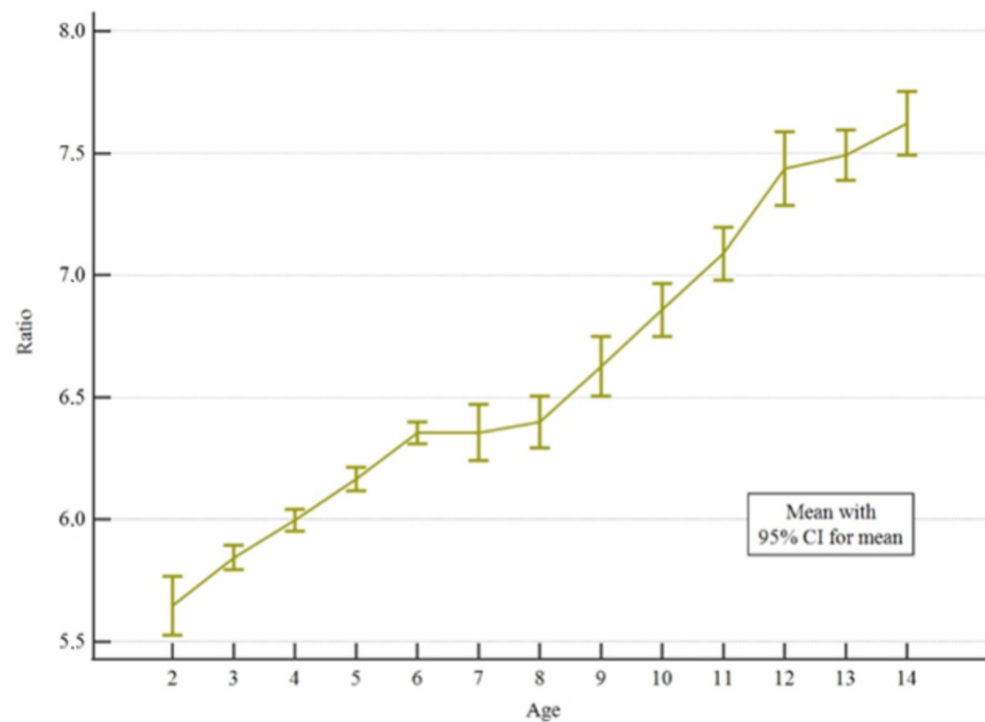


Figure 2. Ratio per age group.

Table 5. Distributions of obtained ratios regarding gender per age group.

Age Group	Gender	N	Mean (SD)	95% CI for the Mean	Median	95% CI for the Median	Min–Max	Skewness	<i>p</i> *
≥2–<3	Male	9	5.77 (0.21)	5.60–5.93	5.72	5.53–5.98	5.50–6.11	0.311	0.05
	Female	10	5.54 (0.24)	5.37–5.71	5.59	5.31–5.78	5.16–5.80	−0.462	
≥3–<4	Male	64	5.76 (0.25)	5.70–5.82	5.80	5.73–5.86	5.19–6.31	−0.412 +	<0.001
	Female	62	5.93 (0.28)	5.86–6.00	5.98	5.89–6.05	5.27–6.43	−0.446	
≥4–<5	Male	94	5.93 (0.31)	5.86–5.99	5.94	5.83–6.02	5.25–6.77	0.052	0.003
	Female	108	6.06 (0.32)	6.00–6.12	6.02	5.96–6.12	5.47–6.79	0.279	
≥5–<6	Male	112	6.10 (0.32)	6.04–6.16	6.07	5.98–6.16	5.45–6.94	0.524	0.001
	Female	88	6.26 (0.35)	6.18–6.33	6.27	6.18–6.33	5.33–7.03	−0.101	
≥6–<7	Male	82	6.25 (0.25)	6.19–6.30	6.26	6.21–6.33	5.66–6.73	−0.267	<0.001
	Female	81	6.47 (0.29)	6.40–6.53	6.46	6.37–6.55	5.83–6.97	−0.116	
≥7–<8	Male	33	6.26 (0.41)	6.11–6.40	6.36	6.11–6.46	5.17–6.99	−0.678	0.03
	Female	22	6.51 (0.39)	6.33–6.68	6.47	6.35–6.67	5.86–7.42	0.365	
≥8–<9	Male	30	6.48 (0.44)	6.32–6.65	6.55	6.24–6.70	5.67–7.32	−0.087	0.08
	Female	25	6.30 (0.32)	6.16–6.43	6.27	6.14–6.41	5.65–7.05	0.428	
≥9–<10	Male	41	6.56 (0.48)	6.41–6.71	6.56	6.38–6.69	5.59–7.72	0.29	0.14
	Female	26	6.74 (0.53)	6.53–6.95	6.78	6.42–7.09	5.84–7.68	−0.013	
≥10–<11	Male	30	6.61 (0.42)	6.46–6.77	6.61	6.43–6.70	5.92–7.35	0.219	0.001
	Female	55	6.99 (0.50)	6.86–7.13	6.95	6.75–7.10	5.79–8.06	0.346	

Table 5. Cont.

Age Group	Gender	N	Mean (SD)	95% CI for the Mean	Median	95% CI for the Median	Min–Max	Skewness	<i>p</i> *
≥11–<12	Male	36	6.96 (0.46)	6.80–7.11	6.97	6.76–7.18	6.00–8.05	0.059	0.02
	Female	38	7.22 (0.44)	7.07–7.36	7.19	7.04–7.41	6.22–8.32	0.189	
≥12–<13	Male	37	7.39 (0.67)	7.16–7.61	7.22	7.00–7.56	6.32–8.95	0.574	0.44
	Female	29	7.50 (0.53)	7.30–7.70	7.46	7.18–7.78	6.64–8.62	0.420	
≥13–<14	Male	34	7.45 (0.45)	7.30–7.61	7.42	7.23–7.69	6.60–8.29	0.002	0.47
	Female	36	7.53 (0.41)	7.39–7.67	7.46	7.30–7.64	6.59–8.60	0.473	
14–<15	Male	38	7.58 (0.51)	7.41–7.75	7.50	7.40–7.77	6.76–8.79	0.321	0.40
	Female	24	7.69 (0.53)	7.47–7.92	7.60	7.42–7.87	6.83–8.90	0.748	

* Student's *t* test; † reject normality (Kolmogorov–Smirnov test).

It is evident that elementary school students (between 10 and 12 years old) experience the most violent psychological and physical changes, and thus, their head canon changes faster (Figure 3). Their proportions change, but very often not at the same time and age period in relation to the observed height.

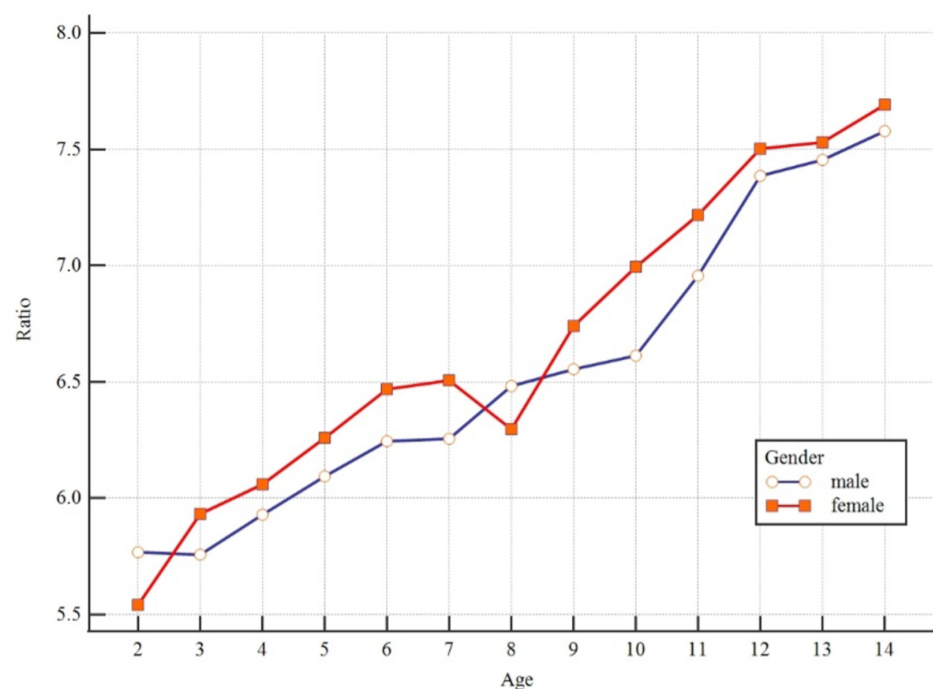


Figure 3. Ratios regarding gender per age group.

4. Discussion

This paper is written from a design and engineering point of view, where existing anthropometric data were used in order to prove canons that could later be used for establishing potential anthropological–biomechanical models and ensure more effective children's product design. The data of head length (HL) to total body height (BH) for this article were extracted from two dissertations of authors who deal with furniture design in educational institutions. The research presented in this paper began with the aim of determining potential head–body canons, and does not present statistical data related to precise anthropometric or medical research. A canon is considered as a set of rules and principles for body proportions as it was set throughout art history. An artistic canon of body proportions (or aesthetic canon of proportion), in the sphere of visual arts, is a formally codified set of criteria deemed mandatory for a particular artistic style of figurative art. Therefore, the goal of this work was to determine a potential canon for children with our available data and highlight the need of future interdisciplinary research.

Considering the previously mentioned researchers of artists and engineers throughout the centuries, it is confirmed that the canons between 5.5 HL and 7.5 HL for children of kindergarten and primary school age are still valid today. It is interesting to note how children's canons, although they move within the observed ratios (5.5 HL to 7.5 HL), have increased over time with regard to the observed age of the children. The made comparison of the canon (Table 6) shows that, for example, according to [25], the canon for children between 4 and 5 years is 5.5, while the new research shows the average value of boys and girls from Table 5 and is 5.99 HL. A similar thing can be seen when comparing, for example, [5] and the results presented in this paper for children 4–5, 10–11, and 14–15 years old.

Table 6. Comparison of canons by different authors through the years.

Autor	[20] Vitruvius (1914) (1st Century BCE)	[24] Richer (1889) [5] Barscay (1989)	[6] Muftić (2001)	Results (2024)
Children's Age	Canon (HL)			
4–5	No data	No data	5.5 HL	5.99 HL
5–6	No data	5.5 HL	No data	6.18 HL
10–11	No data	6.0 HL	No data	6.80 HL
14–15	No data	7.0 HL	No data	7.63 HL
21–35	8 HL	7.5 HL	8 HL	No data

This leads to the conclusion that children of the same age have grown over the last 100 years, which is in line with secular trends (changes in the growth and development of children) confirmed by researchers around the world [15,16,33]. (Table 6)

In addition to the canon of head length (HL) to body height (BH), many researchers observe other body dimensions and proportions in correlation with anthropometry [34], as well as socioeconomic factors [35,36], beauty, growth, and health [37,38], furniture [39], clothes [40], clinical, or nutrition standards [41–43]. For example, the reference charts of sitting height (SH), subischial leg length (SLL), and the sitting height/leg length ratio (SH/LL) are useful tools in assessing body proportion for clinicians and researchers in related areas [44]. Many studies have demonstrated differences between national and WHO reference curves in children older than 5 years [45]. Unfortunately, reference curves for body proportions (SH, SLL, and their ratio) based on state-of-the-art statistics are not available.

Research in China in the nineties of the 20th century [46] found that the arm span to standing height ratio and the upper segment to lower segment ratio are the most useful parameters in describing body proportions. Similar results were confirmed on Japanese children, where the results highlight the usefulness of the Body Proportion Chart method for identifying changes in body proportions during adolescence [47].

It is evident that certain canons also change with age during a child's development towards adulthood. According to the obtained data, it was determined that the changes in the canon are related to all the corresponding harmonic changes of the body in the growth of every healthy child. Applying calculations based on the canons and the harmonic circle could be much simpler than demanding anthropometric measurements, which would speed up the collection of anthropometric data.

The novelty of this research is in the idea of establishing a newer and simpler way of obtaining anthropometric data instead of complex anthropometric measurements and to establish a head–body canon in the future as a measure for creating a potential new anthropological–biomechanical model.

Several important findings emerge from this research that confirm the set hypotheses:

- The existing data established in research [5,6,21,24] on the canon (ratio) of children's head length (HL) to total body height (BH) coincide with the results of our research, which range from 5.59 and 5.72 (2-year-old girls and boys) to 7.50 and 7.60 (15-year-old boys and girls), depending on age and gender.

- Compared to existing data from the literature (Table 6), the canon has increased with regard to the observed age of children, which confirms the secular trends of the past 100 years.
- Observing the growth and development of a child's body in relation to an adult, it is obvious that the corresponding canons change faster with age.
- Although the canons for boys and the girls of the same age are similar, they must be observed separately.
- Studies should consider not only the age and gender, but also the origin, nationality, and other sociodemographic parameters of the child in future research.

We are aware of significant limitations in this research, which are mainly from an anthropometric–medical point of view. Primarily, we used twenty-year-old anthropometric data for school children, simple measuring instruments, and were not focused on anthropometric databases and secular trends. As well, the research was focused on the geographical locations of the Balkan countries where the white Slavic race predominates and where there is very little mixing with other ethnic groups.

As a recommendation for future research, it would be desirable to conduct new anthropometric measurements with 3D- or even 4D-body scanning anthropometry cameras and measuring instruments [48,49], but being aware of usage on young children [50]. Potentially, to involve new growth models and data is also preferable [51].

In the case of new cross-sectional and in-depth anthropometric research in the future, more attention should be paid to checking the accuracy of data through rigorous quality control measures, standardization, reliability levels, and other statistical parameters.

The subject of the canon should be extended to other countries and places around the world, where the sociodemographic factors are different and where additional valuable data can be obtained in children of various ethnic groups. Anthropometric data provide objective physical measurements that can correlate with the aforementioned sociodemographic data. For example, race, color, ethnicity, and other relevant factors could provide a comprehensive understanding of how these variables relate to anthropometric measurements within a study population, since these factors can influence physical characteristics such as height, body mass index (BMI), and other body proportions [36]. Additionally, physical activity levels, dietary habits, medical and family history, and environmental exposures could provide a more comprehensive context for interpreting anthropometric measurements and understanding factors influencing children's physical growth and development and its impact on HL/BH canon. In future research, it would be interesting to clarify and investigate changes in other body proportions (e.g., what is happening with body–limbs, body–thighs, body–limb weight ratio, etc.) and prove other body canons.

All abovementioned details are crucial for exploring potential differences, health outcomes, and cultural influences in the context of anthropometric research. Anthropometric data can be important for a better understanding of biological and health differences within the study population, as well as for evaluating the interaction of these variables with sociodemographic factors in the analysis of study results. All of the above can be connected with the canons, the future modulation of the harmonic models, as well as the creation of a new anthropological–biomechanical model based on the canons.

5. Conclusions

The ratio of the head to the total height (BH/HL) of an adult has been known in the literature since ancient times, but there is little data on the canon of the ratio of the head to the body of preschool and school children.

The obtained results confirm that the canonical changes follow the historical research of artists throughout the centuries, but that they changed according to contemporary secular trends in the growth of children (head-to-body ratio) depending on age and gender.

A calculation based on canons is important for various professions, such as ergonomists, designers, engineers, artists, and others, especially where there is a need to create an anthropological–biomechanical model that could be used to determine other

anthropomeasures depending on gender and age, without undertaking extensive measurements. Creating an anthropological–biomechanical model could significantly simplify the long-term collection of anthropometric data for children’s product design, such as furniture, clothing, toys, playgrounds, and similar products that use children’s anthropometric variables in static and dynamic anthropometry.

Questions related to joining appropriate canons (head–body ratio) still do not have a final form, and further research on this topic is recommended. This study proposes new interdisciplinary research in which new anthropometric data of children with a larger sample from different sociodemographic groups will be taken, using modern measuring instruments and appropriate statistical processing, to confirm the validity of the children’s canon in the future. It is desirable to find other body proportions in relation to body height in order to be able to create a more comprehensive anthropological–biomechanical model based on the canon and the harmonic model.

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Institutional Review Board Statement: All research with children in kindergartens and elementary schools are part of larger studies whose data are published in the authors’ doctoral theses [11,12] and have undergone the process of ap-proval for the conduct of research by the relevant institutions. Permissions to enter kindergartens and schools to conduct research were obtained in all locations/cities from the Ministries of education and sport in Croatia, individual state or city offices in Croatia, N. Macedonia, and Bulgaria, as well as from kindergarten and school principals, and parents/guardians whose children participated in the research. Additionally, for the research in kindergarten polygons in Zagreb (Croatia), the Ethics Committee of the University of Zagreb approved and confirmed that the research complies with the principles and rules of the Code of Ethics [32], Class: 602-04/19-17/11, No. 308-106/334-19-6.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: Detailed data are available in doctoral theses [11,12] and published reports [33]. Additional information is available through a direct request to the corresponding author.

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Conflicts of Interest: The authors declare no conflicts of interest.

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